

POTENTIOMETRIC TITRATIONS

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In a potentiometric titration, a suitable electrode immersed in the solution to be titrated acts as the 'indicator'. The indicator electrode is paired with a reference electrode and the two electrodes are connected to an electronic voltmeter. The emf of the indicator electrode changes gradually with the change of concentration of ions caused by the addition of titrant from the burette. The equivalence point is indicated by a sharp change in electrode potential.

Since the reference electrode potential has a constant value, any change in the indicator electrode potential is reflected by a similar change in the cell potential. Therefore, the equivalence point can be found by plotting a graph between the cell emf and the volume of titrant added from the burette. A sharp rise of the curve shows the equivalence point and the corresponding volume on the graph is the volume of the solution used for the titration.

Principle

Emf of a cell depends on the concentration of the electrolytes with which the electrodes are in contact. Therefore, the electrode reaction is,



As the concentration of M^{n+} changes, the emf of the cell also changes correspondingly.

Thus, the potentiometric titrations involve the measurement of emf between reference electrode and an indicator electrode, with the addition of the titrant.

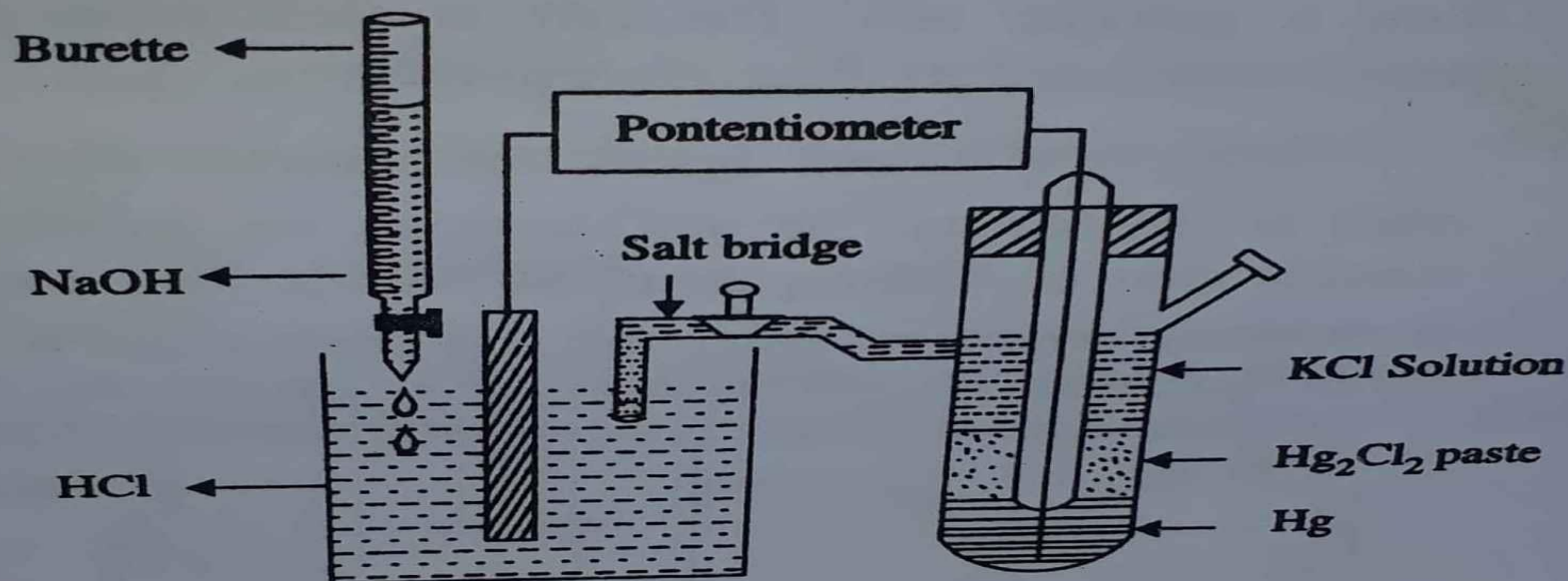


Fig. 2.8 Potentiometric titration

2.10.1 *Types of potentiometric titration*

Potentiometric titrations fall into the following three categories.

1. Acid-base titrations.
2. Redox titrations.
3. Precipitation titrations.

The potentiometric titrations may be of three types :

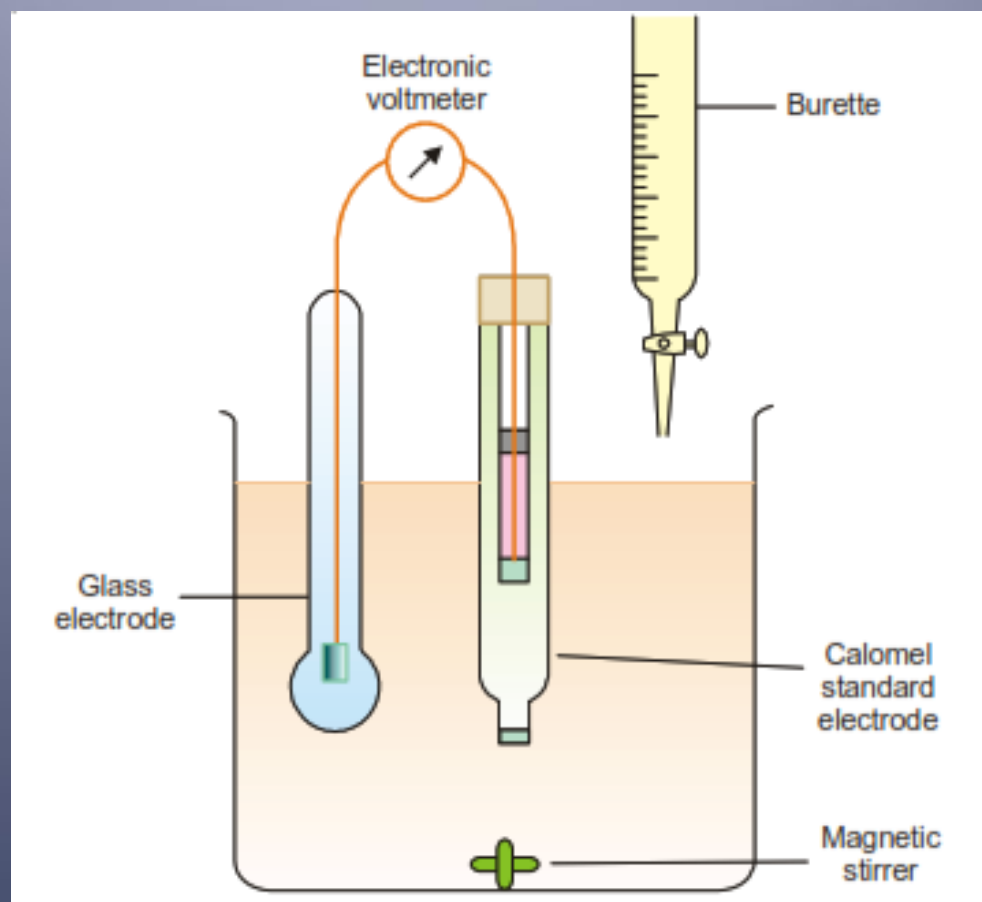
(a) Acid-base titrations

(b) Oxidation-reduction titrations

(c) Precipitation titrations

Acid-base Titrations

The apparatus used for potentiometric acid-base titrations is shown in Fig.

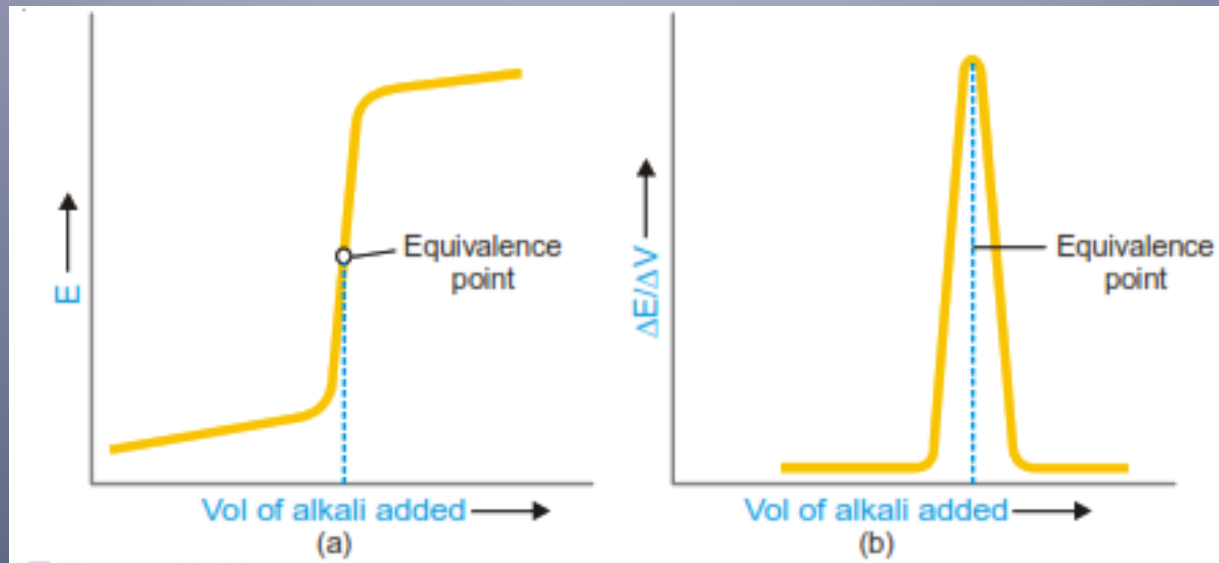


A hydrogen electrode or a glass electrode is immersed in solution of the acid whose strength is to be determined. The glass electrode is coupled with a standard calomel electrode. The cell thus formed is connected to the potentiometer or electronic voltmeter. When alkali is added, pH of the solution changes. The emf of the cell also changes with pH of the solution in accordance with the relation.

$$E = E^{\circ} + 0.0591 \text{ pH}$$

The standard alkali solution is then added from the burette in small volumes. After each addition, the emf of the cell is recorded. The emf is then plotted against the volume of alkali added. The shape of the curve for the titration of a strong acid against strong alkali (HCl versus NaOH) is shown in Fig (a). The steepest portion of the curve indicates the equivalence point. However, when the solutions are very dilute, or weak acids or bases, are involved, the steepness of the curve is less marked and it is difficult to judge the end-point. In such a case, we plot the slope of the curve, $\Delta E/\Delta V$ against the volume of alkali used. The maximum of the curve indicates the end-point.

Potentiometric titration curve of an acid and a base



1. Acid-base titrations (HCl vs NaOH)

The HCl solution is taken in a beaker and the indicator electrode (glass electrode) is placed in it. It is then connected to a reference electrode, (calomel electrode) to form a galvanic cell. The cell is then connected to the potentiometer and its E_{cell} is determined.

When it is titrated against the standard NaOH solution, taken in the burette, the emf is going on ~~decreasing~~ *increasing* as the H^+ ion concentration decreases.



At the end point, emf is suddenly ~~decreases~~ *increased*. After the end point there is no noticeable change in the potential.

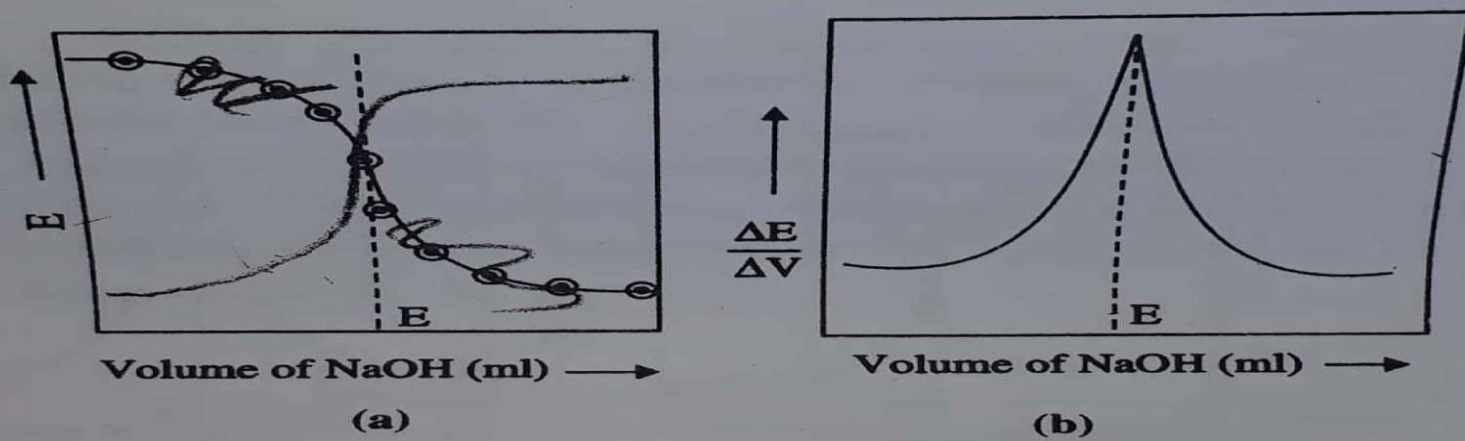


Fig. 2.9 E_{cell} Vs Vol. of NaOH

When the emf is plotted against the volume of NaOH added, a curve of the type shown in figure 2.9 is obtained. The end point is the point, where the slope of the curve is maximum.

A more sensitive and satisfactory method of detecting the end point will be the graph of $\frac{\Delta E}{\Delta V}$ against volume of NaOH. The resulting curve rises to a maximum at the equivalent point, which is the end point.

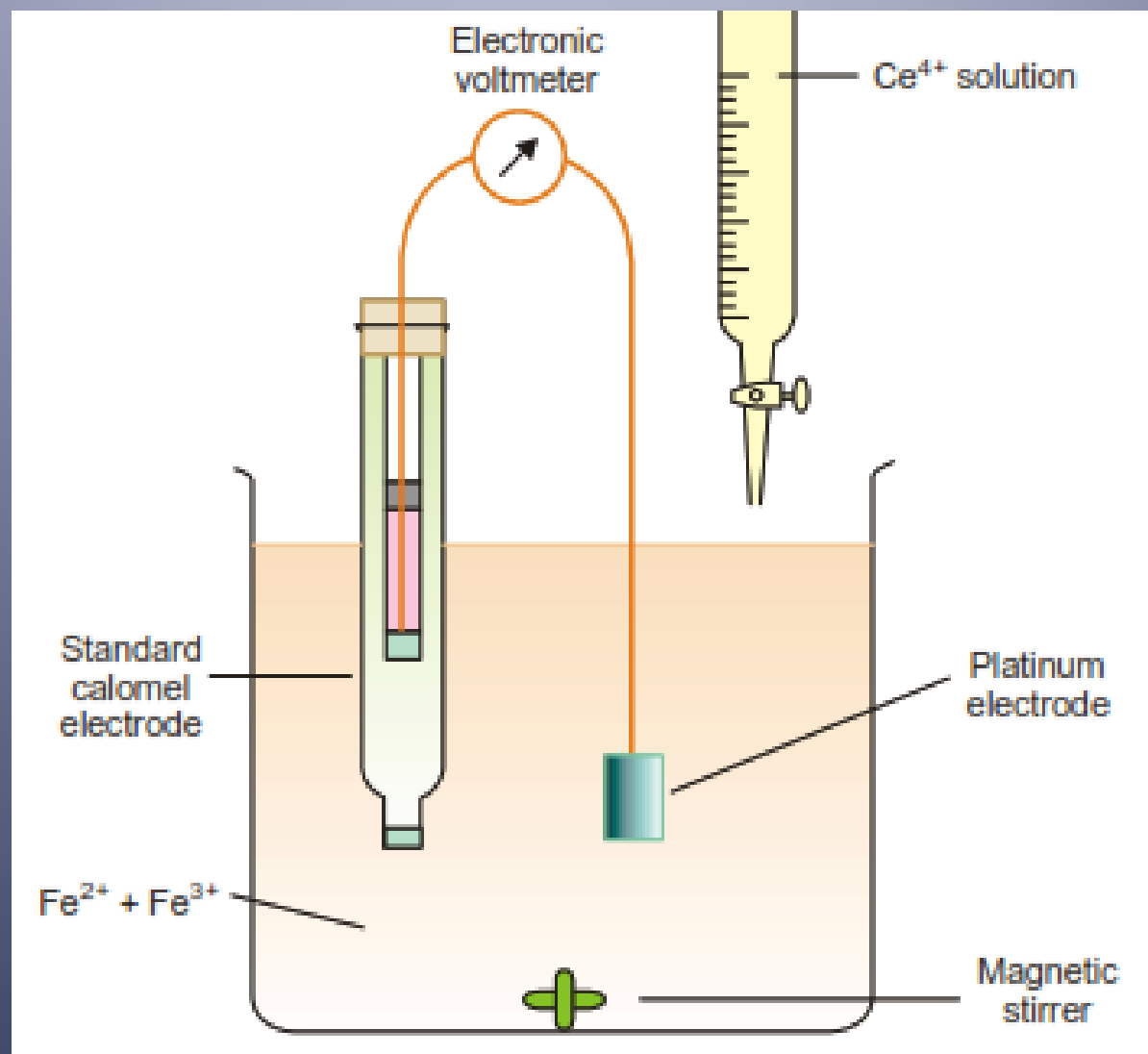
Oxidation-reduction Titrations

The titration of ferrous ions (Fe^{2+}) with ceric ions (Ce^{4+}) is an example of oxidation-reduction (or redox) titration. Fe^{2+} ion is oxidised to Fe^{3+} ion, while Ce^{4+} is reduced to Ce^{3+} ion.



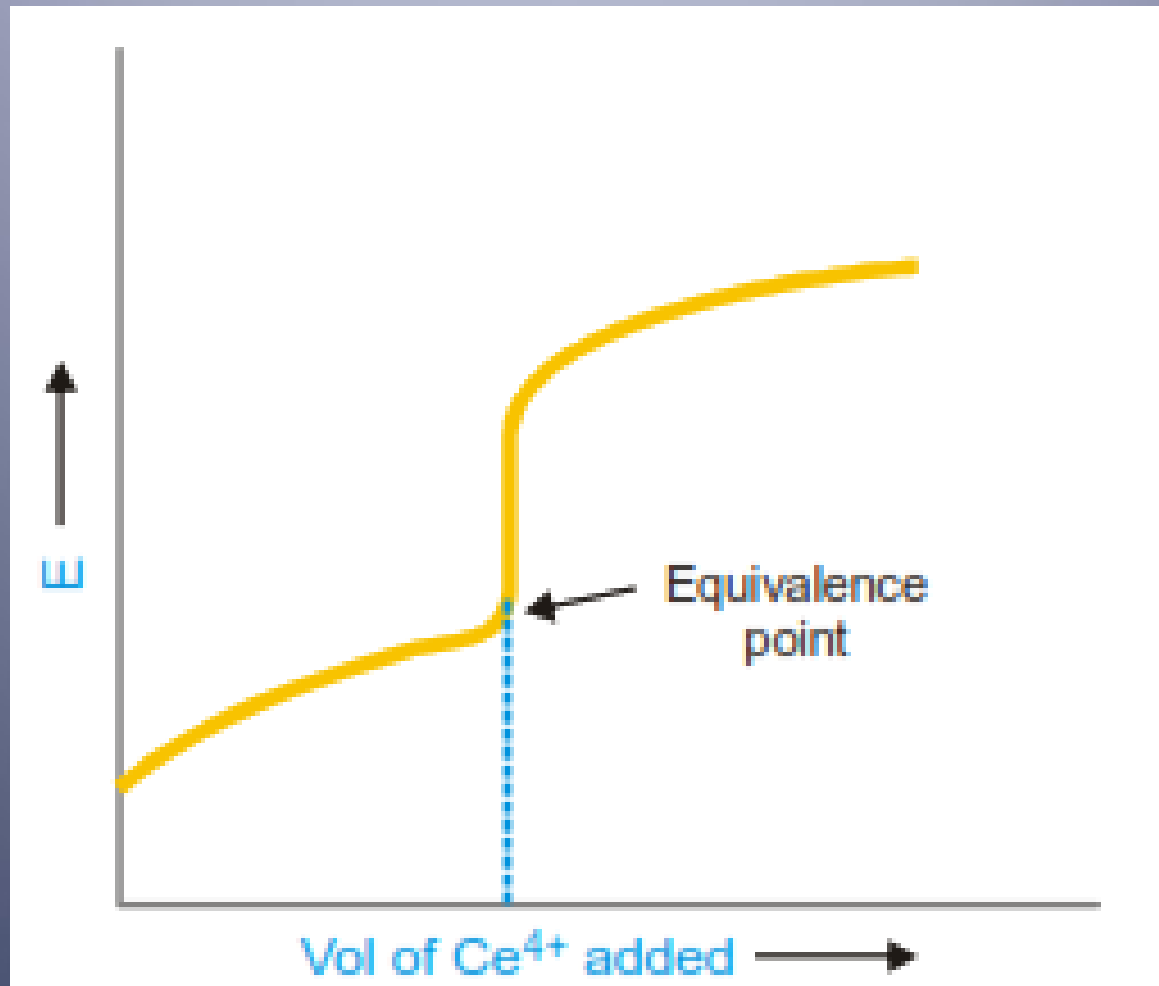
This titration can be carried in the apparatus shown in Fig.. The indicator electrode is a shiny platinum strip dipping in the solution of Fe^{2+} ions, and it is connected to a standard calomel electrode. The Ce^{4+} solution is added from the burette and the cell potential, E , recorded after each addition.

Apparatus for potentiometric titration of Fe^{2+} with Ce^{4+}



The potential of the platinum electrode depends on the ratio $[\text{Fe}^{3+}]/[\text{Fe}^{2+}]$. The potential of the cell, E , also changes with the change of the ratio $[\text{Fe}^{3+}]/[\text{Fe}^{2+}]$. Therefore, the cell potential changes with the addition of Ce^{4+} ions from the burette. Fig. shows how the potential of the cell changes during the titration. At the equivalence point there is a sharp rise of potential which indicates the endpoint. Potentiometric titrations of this type are particularly useful for coloured solutions in which an indicator cannot be employed.

Potentiometric titration curve of Fe^{2+} ions and Ce^{4+} ions.



3. Redox titration (FeSO_4 vs KMnO_4)

The KMnO_4 solution is taken in a ~~beaker~~ ^{burette} and the indicator electrode (platinum electrode) is inserted in it. It is then connected to a reference electrode (calomel electrode), to form a galvanic cell. The cell is then connected to the potentiometer and its E_{cell} is determined.

When it is titrated against the standard FeSO_4 solution, taken in the burette, the emf is going on ~~decreasing~~ ^{increasing} as the concentration of KMnO_4 decreases. At the end point the emf is suddenly decreases. After the end point there is no change in the potential.

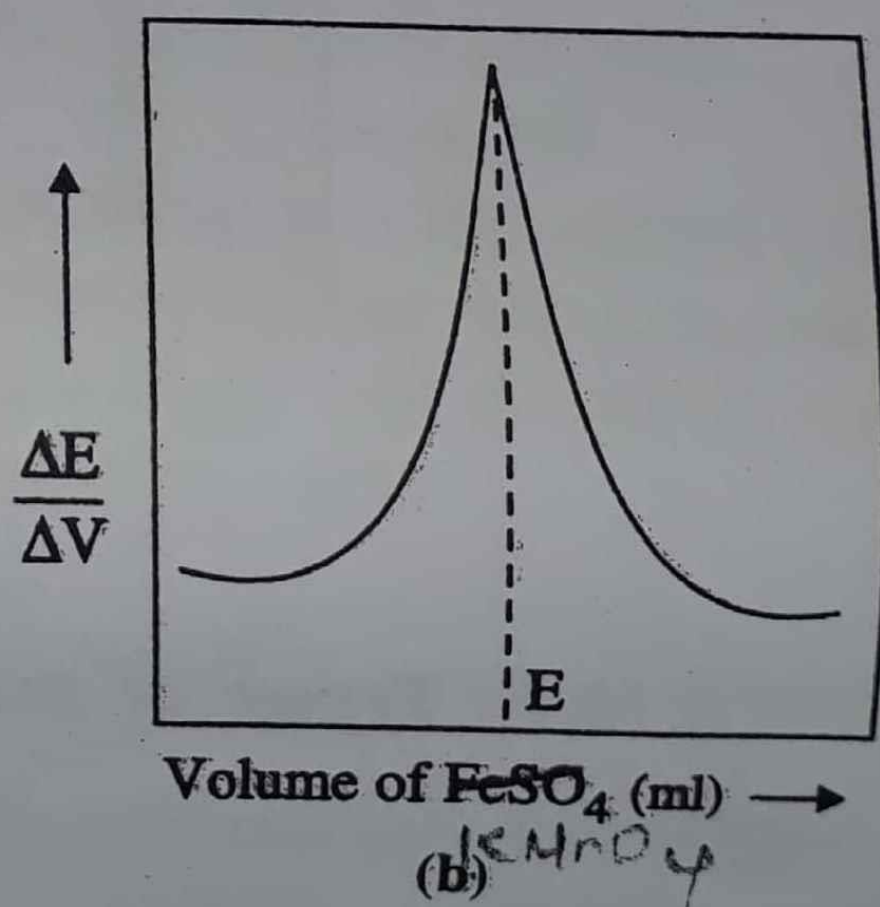
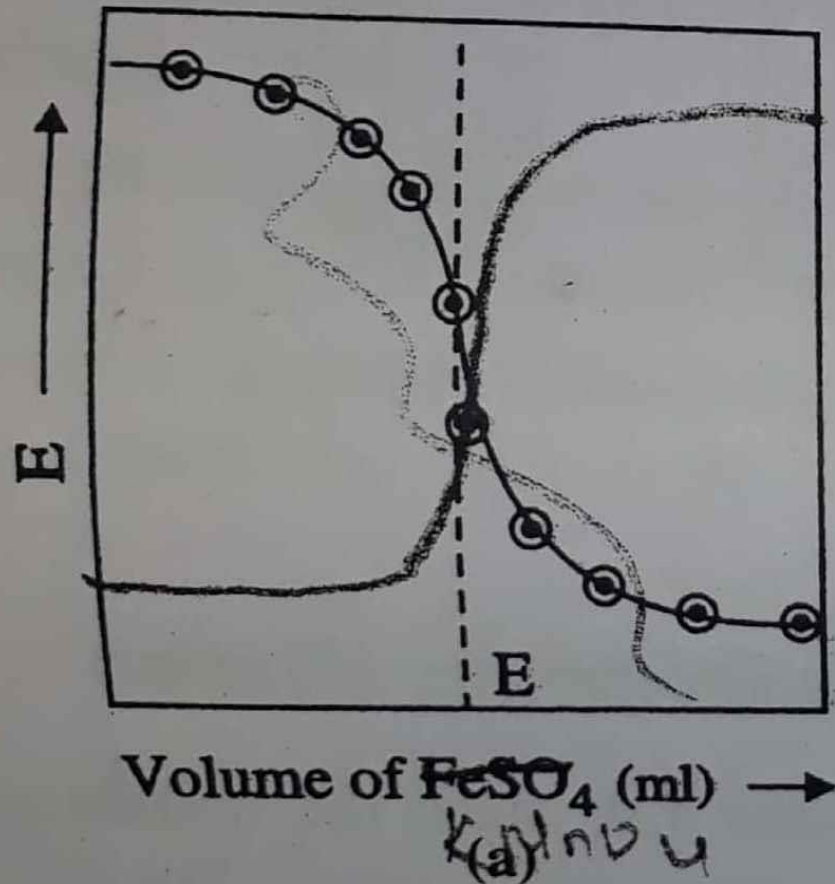


Fig. 2.11 E_{cell} Vs Vol. of FeSO_4

The curve for E vs Vol of FeSO_4 and $\frac{\Delta E}{\Delta V}$ vs Vol. of FeSO_4 (Fig. 2.11) are exactly similar to the curve obtained in the above methods.

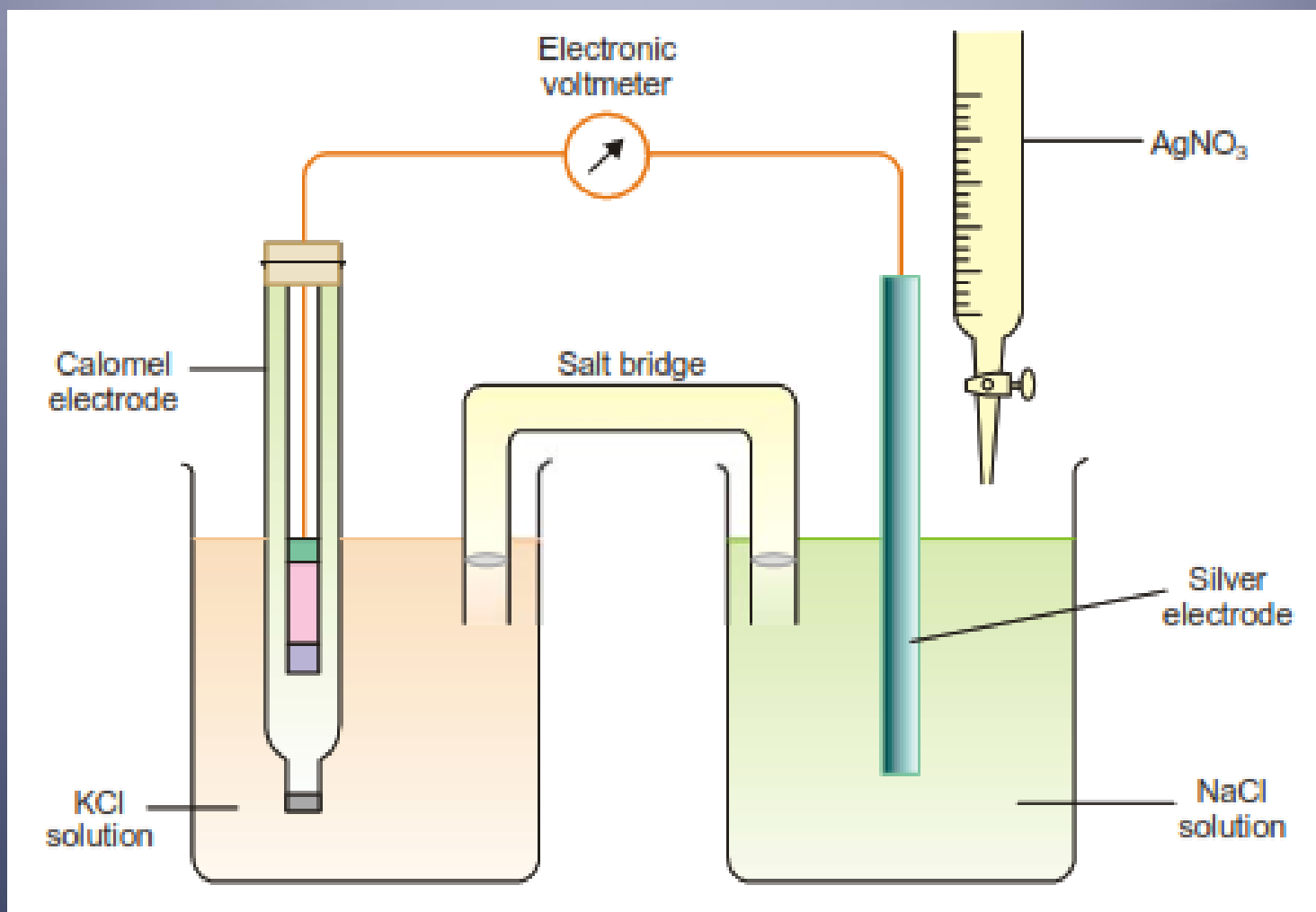
Precipitation Titration

A typical precipitation titration is that of sodium chloride solution against silver nitrate solution. The apparatus set up for the purpose is shown in Fig. A silver electrode dipping in the unknown sodium chloride solution is coupled with a calomel electrode through a salt bridge. However, if the calomel electrode were in direct contact with a solution containing excess silver ions, chloride would seep through the sintered base and react to form an insoluble layer of silver chloride. Any change in the cell potential is due to changes in concentration of Ag^+ ions around the silver electrode.

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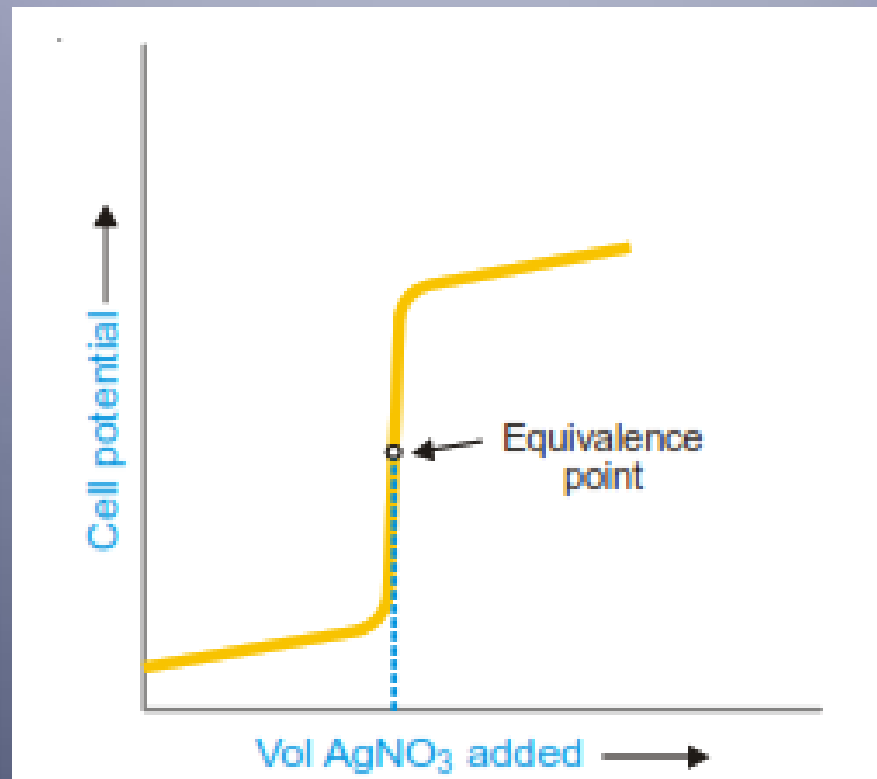
Potentiometric titration of sodium chloride against silver nitrate solution



Initially the concentration of Ag^+ ions will be zero. But as silver nitrate is added from the burette, silver chloride is precipitated. Now the solution will contain a small concentration of Ag^+ ions formed by the slight dissociation of silver chloride. This concentration will increase slightly as Cl^- ions are removed in order to maintain the solubility product $K_{\text{sp}} = [\text{Ag}][\text{Cl}^-]$. After the equivalence point, the concentration of Ag^+ ions and, therefore the silver electrode potential will rise very sharply owing to the presence of excess of Ag^+ ions. The volume of AgNO_3 solution used to reach the equivalence point as shown in Fig.

+

Potentiometric titration curve.

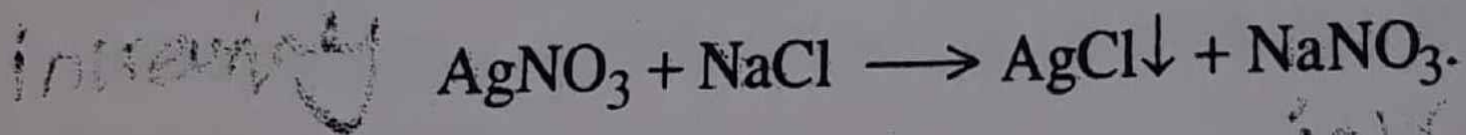


2. Precipitation titration (AgNO_3 vs NaCl)

The AgNO_3 solution is taken in a beaker and the indicator electrode (silver-silver ion electrode) is placed in the

beaker. It is then connected to a reference electrode (calomel electrode), to form a galvanic cell. The cell is then connected to the potentiometer and its E_{cell} is determined.

When the AgNO_3 solution is titrated against the standard NaCl solution, taken in the burette, the emf is going on decreasing as the Ag^+ ion concentration decreases.



intensity

At the end point, emf is suddenly decreases. After the end point there is no noticeable change in the potential.

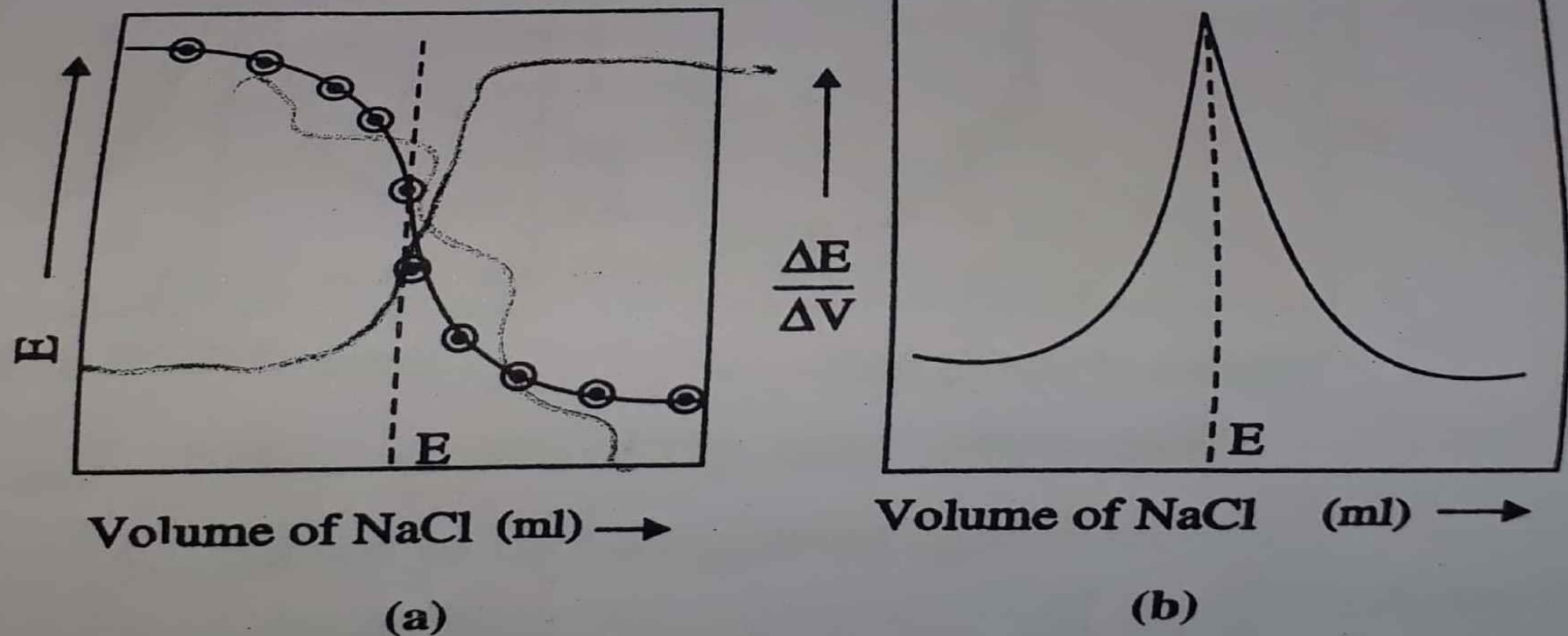


Fig. 2.10 E_{cell} Vs Vol. of NaCl

When the emf is plotted against the volume of NaCl added a curve of the type shown in figure 2.10 is obtained. The end point is the point, where the slope of the curve is maximum.

The graph $\frac{\Delta E}{\Delta V}$ vs vol. of NaCl is plotted as above, to get the accurate end point.

Advantages of Potentiometric Titrations

Advantages of potentiometric titrations

1. The necessary apparatus required is cheap and easily available.
2. This method can be used for coloured solution.
3. Fixing up end point is easier when compared to the titrations in which indicators are used to fix up end points.
4. Very dilute solutions can be titrated with accuracy.
5. Several components may be titrated in the same solution.

WHAT IS A BUFFER SOLUTION ?

It is often necessary to maintain a certain pH of a solution in laboratory and industrial processes. This is achieved with the help of buffer solutions, buffer systems or simply buffers.

A buffer solution is one which maintains its pH fairly constant even upon the addition of small amounts of acid or base.

In other words, a buffer solution resists (or buffers) a change in its pH. That is, we can add a small amount of an acid or base to a buffer solution and the pH will change very little.

Types

Two common types of buffer solutions are :

(1) a weak acid together with a salt of the same acid with a strong base. These are called Acid buffers e.g., $\text{CH}_3\text{COOH} + \text{CH}_3\text{COONa}$.

(2) a weak base and its salt with a strong acid. These are called Basic buffers. e.g., $\text{NH}_4\text{OH} + \text{NH}_4\text{Cl}$.

Buffer solution

Pure water is neutral and it has pH value of 7. If a small amount of HCl is added, the pH value becomes less than 7 and if a small amount of NaOH is added the pH becomes greater than 7. If we take a mixture of acetic acid and sodium acetate, the pH of the mixture does not change considerably even after the addition of small amount of HCl or NaOH. Such a mixture whose pH remains constant even after the addition of small amount of acid or base is called buffer solution.

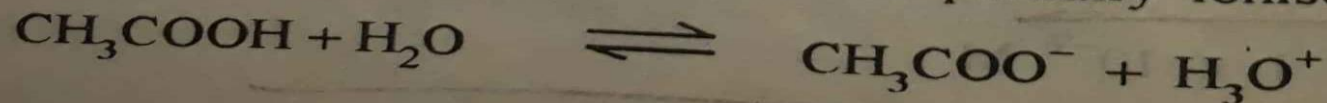
Generally, buffer solutions are

(1) Weak acid and its salt (e.g., acetic acid and sodium acetate). This type of buffer solutions are acidic buffers.

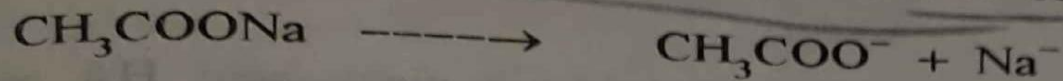
(2) Weak base and its salt. (e.g. ammonium hydroxide and ammonium chloride). This type of buffer solutions are called basic buffers.

Buffer action: The reactions responsible for maintaining constant pH after the addition of acid or base in buffer solutions are called buffer action. The buffer action of acetic acid and sodium acetate mixture can be explained as follows.

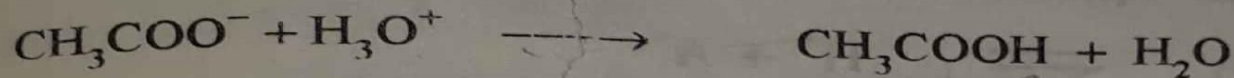
Acetic acid is a weak acid and is partially ionised in solution.



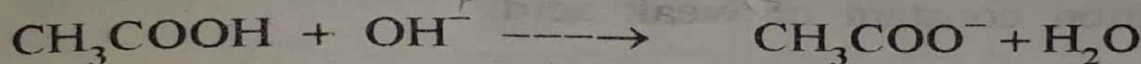
Sodium acetate is a strong electrolyte and is completely ionised.



when a small amount of acid is added to this mixture, H_3O^+ ions of this acid combines with equal number of acetate ions to form acetic acid molecules. Therefore, the net $[\text{H}_3\text{O}^+]$ and pH remain constant.

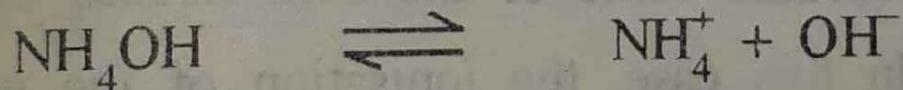


when a small amount of base is added, the OH^- ions of the base neutralise acetic acid molecules to form sodium acetate. Therefore the net $[\text{H}_3\text{O}^+]$ and pH remain constant.

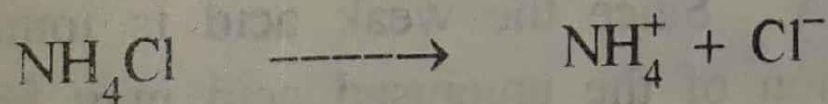


In the case of ammonium hydroxide-ammonium chloride buffer mixture, the buffer action is as follows,

Ammonium hydroxide being a weak base is partially ionised



Ammonium chloride is a strong electrolyte and is completely ionised (X)

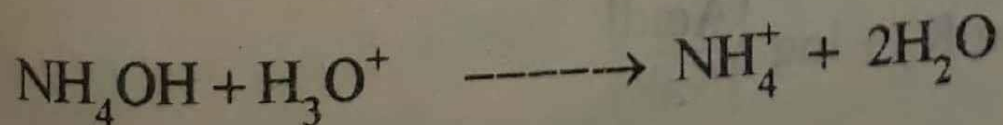


when a small amount of base is added to the mixture, the OH^- ions of the base combine with equal number of NH_4^+ ions to give NH_4OH .



Therefore, the $[\text{OH}^-]$, pOH and pH of the solution remain constant.

When a small amount of acid is added to the mixture, the H_3O^+ ions of the acid are neutralised by NH_4OH to form NH_4^+ and water.



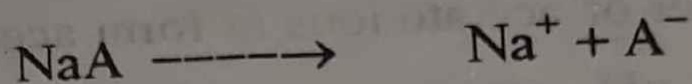
Therefore, the $[\text{OH}^-]$, pOH and pH of the solution remain constant.

Calculation of pH of a buffer solution

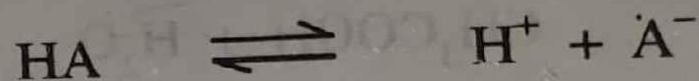
Calculation of pH of buffer solution:

Henderson equation

Consider a buffer mixture containing a weak acid HA and its salt NaA. Since the salt is a strong electrolyte it is completely ionised.



The ionisation equilibrium of the weak acid is represented as



Applying the law of mass action to the above ionisation equilibrium

$$K_a = \frac{[\text{H}^+][\text{A}^-]}{[\text{HA}]}$$

where K_a is the ionisation constant of weak acid.

$$\therefore [\text{H}^+] = K_a \frac{[\text{HA}]}{[\text{A}^-]}$$

Strictly speaking, the concentration of the acid $[HA]$ in this equation indicates the actual concentration of the acid. And this is slightly less than the original concentration because some of the acid ionises into its anion A^- and hydrogen ions. In this case, the ionisation of the weak acid is further suppressed by the presence of the common ion namely A^- ions produced from the salt NaA . Since the weak acid is ionised to a very small extent, the concentration of the unionised acid may be taken as the original concentration of the acid. Actually the concentration of A^- ions is slightly more than the original concentration of the salt due to the formation of A^- ions to a very small extent by the ionisation of the acid. But this correction is small and is usually omitted. Hence, the concentration of A^- ions is taken as the original concentration of the salt itself.

$$\therefore [H^+] = K_a \frac{[Acid]}{[Salt]}$$

Upon taking logarithms of the above equation

$$\log [H^+] = \log K_a + \log \frac{[Acid]}{[Salt]}$$

Changing the sign throughout, the above equation can be written as

$$-\log[H^+] = -\log K_a - \log \frac{[\text{Acid}]}{[\text{Salt}]} \quad \dots (23)$$

$$\text{or } -\log[H^+] = -\log K_a + \log \frac{[\text{Salt}]}{[\text{Acid}]} \quad \dots (24)$$

But, from definitions

$$-\log[H^+] = \text{pH}$$

$$-\log K_a = \text{pK}_a$$

$$\therefore \text{pH} = \text{pK}_a + \log \frac{[\text{Salt}]}{[\text{Acid}]} \quad \dots (25)$$

Similarly, it can be shown that for a mixture of weak base and its

salt.

$$\text{pOH} = \text{pK}_b + \log \frac{[\text{Salt}]}{[\text{Base}]} \quad \dots (26)$$

where

$$\text{pOH} = -\log [\text{OH}^-]$$

$$\text{pK}_b = -\log K_b$$

Equations (25) and (26) are referred as Henderson equations.

Application of Buffer Solutions

Applications of buffer solutions:

The control of pH by buffers is important in living organisms the processing of food. The buffering action of blood is due to the presence of weak acid, carbonic acid and its salt, sodium bicarbonate. The pH of stomach fluid is 1.6.

pH plays an important role in almost every industrial process. It has been found that the success of several industrial processes is largely dependent on the control of pH of the system. Some other applications of buffer solutions are given below :

1. Manufacture and refining of sugar.
2. Purification of water.
3. Manufacture of leather
4. Manufacture of paper
5. Manufacture of milk products

Decomposition Potential (or) Decomposition Voltage.

(90)

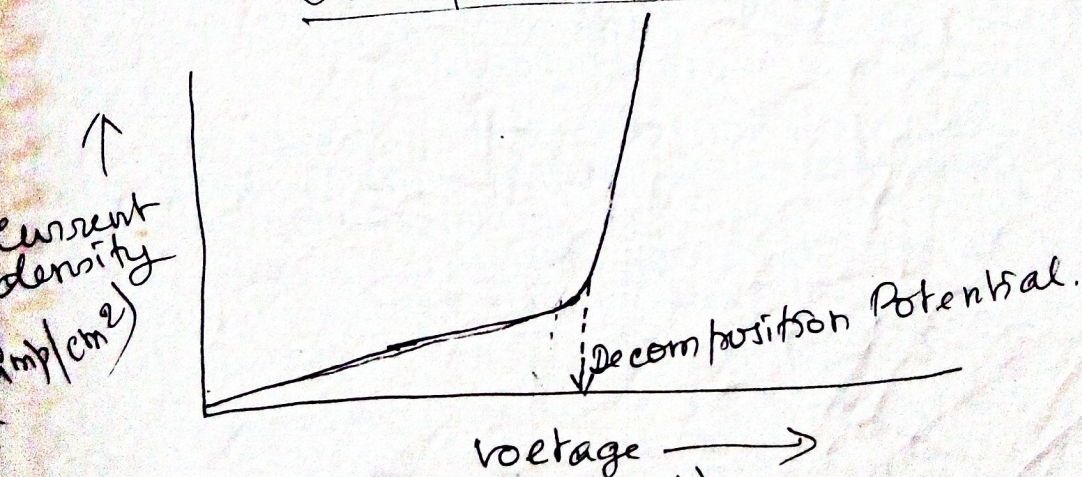
The minimum voltage which is sufficient to overcome the back e.m.f. is called the Decomposition Potential of the solution.

It is known that the products of electrolysis exert a back e.m.f. which counteracts the applied primary e.m.f. Thus electrolysis of a compound can not be carried indefinitely unless the back e.m.f. is overcome.

For eg. when an external e.m.f. of one volt is applied to 2 Platinum electrodes dipping in Dil. H_2SO_4 , electrolysis of H_2O , takes place. But, very soon, the electrolysis comes to a stop due to the back e.m.f. i.e. the back e.m.f. becomes greater than the applied e.m.f.

If the applied e.m.f. is gradually increased, a point is reached when the electrolysis again continues and does not stop even though the applied e.m.f. is not increased. Thus a certain minimum voltage must be applied for a continuous electrolysis to take place.

This minimum voltage which should be applied to the solution to overcome the back e.m.f. due to polarisation is known as Decomposition Potential.

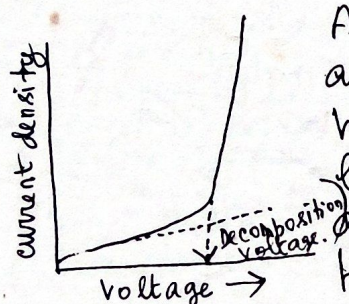


The plot of current density vs the cell voltage is indicated above. The voltage at which there is a sudden increase in current density is the decomposition potential.

Importance and significance of decomposition voltage.
 The decomposition potential varies from solution to solution. During the electrolysis, the metal ions travel towards the cathode. If the metal has a higher solution pressure, it will try to send these ions back into the solution. i.e. the solution will have high decomposition potential. Thus a metal with a high solution pressure (for eg. Zn) requires a high emf, to be deposited on the cathode than a metal with a low solution pressure (for eg. Cu).

This principle is used in Refining and Electroplating of metals. For eg. the decomposition potentials of CuSO_4 and ZnSO_4 with Cu and Zn electrodes ^{are} 1.5 and 2.5 volts respectively. When the applied emf is less than 2.5 volts between the Cu electrodes, Cu alone will be deposited while Zn remains in solution. Thus when Cu^{++} ions are completely deposited and removed from the solution the successive deposition of Zn^{++} ions can be done by increasing the applied e.m.f.

Measurement of decomposition potential.



A graph of current density against the applied voltage is plotted. The applied voltage is gradually increased until bubbles of gas or deposition of metal on an electrode can be detected with the help of a microscope.

- Note.
- 1) Decomposition potential vary with the conditions but it is characteristic of definite electrolytes. Thus the decomposition potential at CuSO_4 is 1.6 volt, $\text{Pb(NO}_3)_2$ is 1.8 volt and $\text{Cd(NO}_3)_2$ is 2.3V.
 - 2) Most of the strong acids and bases have decomposition potential of 1.7 volts. They all behave alike in that at 1.7 volts or above they give off H_2 and O_2 at the electrodes.
 - 3) Acids, like con. HCl which give products other than H_2 and O_2 have decomposition voltages less than 1.7 volts.

Q.4 - Apr, 1986 - Define decomposition potential. How is this experimentally determined? (6 marks)

Overvoltage or Overpotential.

The difference between the voltage at which a gas is actually evolved during electrolysis and the theoretical value at which it is expected to be evolved is known as overvoltage.

For eg. the electrode potential of Zn in contact with one molar (1M) $ZnSO_4$ solution is 0.76 volt. i.e. its theoretical decomposition voltage is 0.76 volt. But, in practice, it is observed that the actual decomposition potential required to deposit Zn^{++} ions on Zn cathode is more than 0.76 volt. This difference between the applied voltage and the theoretical voltage is known as overvoltage or overpotential.

Overvoltage for deposition of metals on metal electrodes are usually low and they can be ignored.

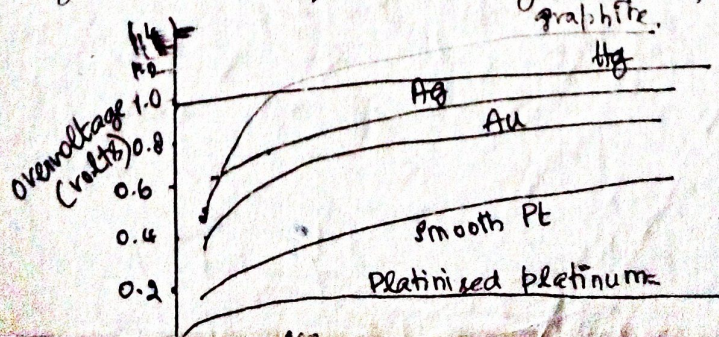
But, overvoltage for evolution of gases have usually significant values.

Hydrogen overvoltage or Bubble overvoltage.

The difference between the potential of the electrode at which hydrogen gas is actually evolved and the theoretical value at which it is expected to be evolved (in the same solution) is known as hydrogen overvoltage or bubble overvoltage.

(Similarly Oxygen overvoltage is defined in the same manner)

On platinised-platinum and at zero current density the overvoltage of hydrogen is zero since these are the conditions under which the hydrogen-electrode is perfectly reversible. However as the current density increases the hydrogen overvoltage also increases. (In fact its value is 0.05 volt at $25^\circ C$ at a current density of 1.5 ampere per sq. cm. While at other cathodic surfaces the overvoltage is much higher as may be seen from the graph).



Hydrogen overvoltage on various metals in 1N. H_2SO_4 at $25^\circ C$

current density = current per unit area

It is clear from the graph that the overvoltage of hydrogen is low only on platinised-platinum but even on ~~smooth~~ smooth platinum surfaces it is low at very small current densities and increases rapidly with increasing current density to approach a limiting value.

Gold, graphite, tellurium and palladium cathodes also behave in the similar way.

However at zero current density, the hydrogen overvoltage on Ag, Bi, Sn, Fe, Ni, Cd and Pb surfaces is already quite large. These electrodes, therefore, can not act reversibly under any condition and it is impossible to deposit hydrogen on them at anywhere near the reversible potentials.

The hydrogen-overvoltages for a few metals are as follows.

Metal.	Mg	Pb	Zn	Cu	Fe	Ni	Au	Platinised-platinum
hydrogen overvoltage in 1M HCl at 0.001 amp/cm ²	1.04	0.88	0.72	0.54	0.4	0.3	0.17	0 (volt)

The hydrogen overvoltage increases as the catalytic capacity of the metal for recombination ~~of~~ between H-atoms decreases.

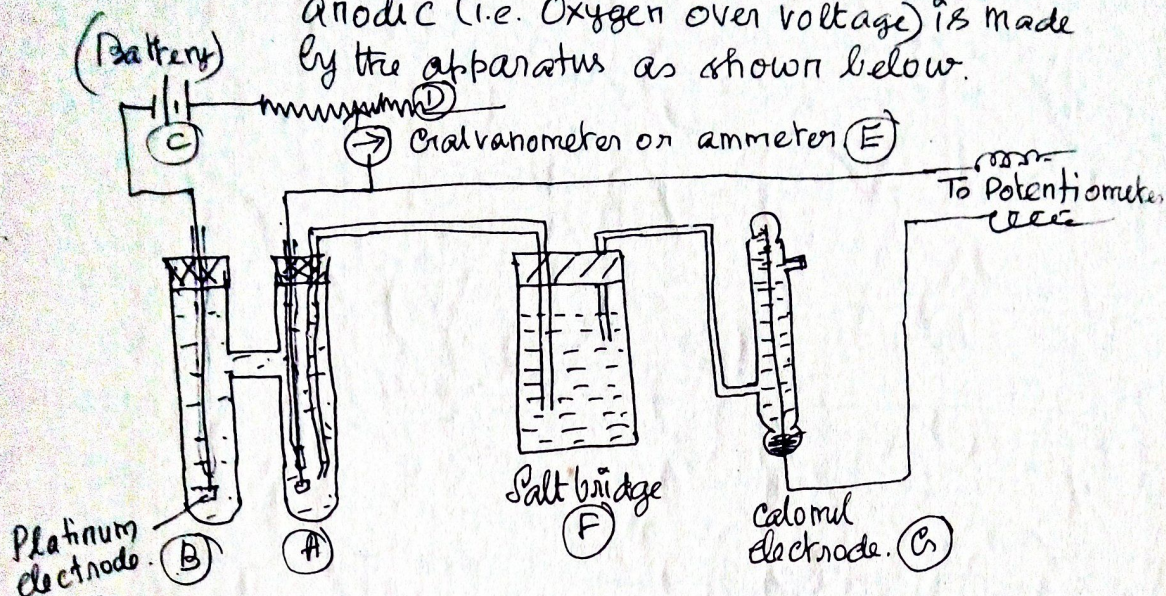
Measurement of hydrogen-over-voltage.

For the measurement of hydrogen over voltage the solution should be free from O₂ or other such ~~reducible~~ materials. Hence, the solution is generally saturated with H₂ gas at one atmosphere.

Then a current of definite strength is passed through the solution for a sufficient time until the potential (i.e. the applied voltage) becomes constant. The cathode potential is then measured by combining it with a reference electrode. If a std-H₂-electrode is used as a reference electrode, the difference of potential between the std-H₂-electrode and Cathode gives directly the value of over voltage.

Determination of overvoltage.

The measurement of over voltage either Cathodic (i.e. hydrogen over voltage) or anodic (i.e. Oxygen over voltage) is made by the apparatus as shown below.



The electrolysis vessel is a H-shaped cell containing the experimental solution with the electrodes of known cross sectional area. These are connected in series with an external battery through a variable resistance. The current flowing through the electrolytic cell is indicated on the galvanometer or ammeter.

The potential of the experimental electrode during electrolysis can be determined by connecting it through a salt bridge (F) to a reference electrode (G) i.e. a calomel electrode.

The emf of the cell composed of the electrodes (A) and (G) is determined with the help of a potentiometer. The potential of (A) is then calculated by subtracting the known potential of the reference electrode from the emf of the cell.

By adjusting the resistance (D) a series of emf readings of (A) can be obtained for various values of the current passing through the cell.

The current density is obtained by dividing the current strength at each point by the area of the electrode (A). The emf of (A) at each point is obtained by subtracting the potential of the reference electrode (calomel electrode) from the measured emf of the cell.

Finally the overvoltage (either cathodic or anodic) can be obtained by subtracting the reversible potential of (A) from the actual (observed)

Factors affecting over voltage.

- ① Influence of pH on overvoltage. At present, the available data show that the composition and the H^+ ion concentration, i.e. pH of the solution have no effect on hydrogen overvoltage. However, strongly acidic or alkaline solutions, sometimes deviations take place and these may be due to the large concentration of H^+ ions or OH^- ions.
- ② Influence of temperature: Since overvoltage is due to a slow stage in the ionic discharge process it is evident that increase of temperature will decrease overvoltage. Generally the change is approximately 2 millivolts per degree, and it is higher at higher temperatures.
- ③ Influence of pressure. The overvoltage is very slightly affected at higher pressures. But, at lower pressures the overvoltage rapidly increases on Cu, Ni and Hg cathodes.
- ④ Influence of surface. An increase in the effective area of the surface causes a decrease in overvoltage. The overvoltage is greater on smooth and polished surface than on rough or pitted surfaces.
The Hydrogen overvoltage of ~~smooth~~ platinum in dil. H_2SO_4 is 0.04 volt while it is ~~0.04~~ 0.05 volt for smooth platinum (which is ~~coarser~~ coarser).
- ⑤ Influence of current density.

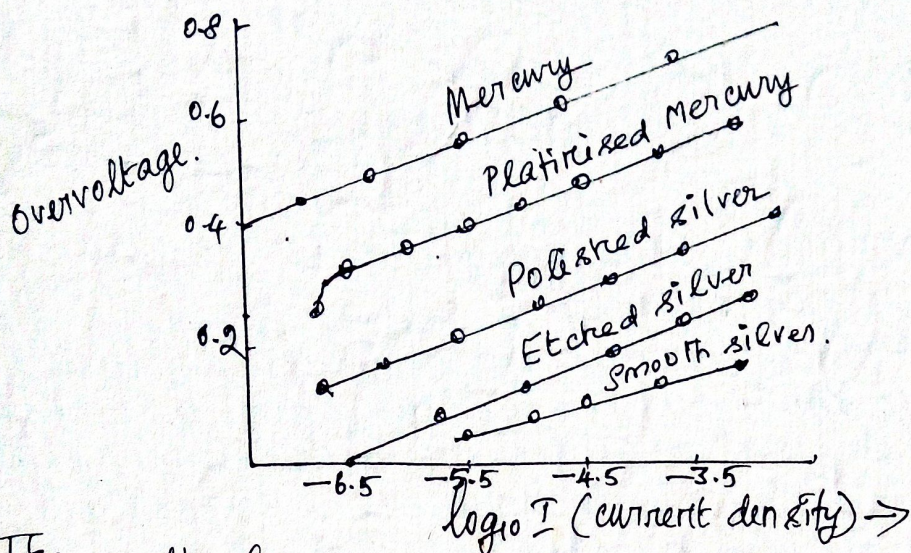
Provided the conditions are such that the hydrogen is not removed by reaction with O_2 or other oxidising agent, or by diffusion away from the cathode, the variation of hydrogen overvoltage (or oxygen overvoltage) with current density at constant temperature can be represented by the equation

$$\eta = A + B \log_{10} I$$

Where, η ... is overvoltage
 I ... is current density.
 A and B are constants.

The above relationship was found to be applicable for a number of cathodes at current densities varying

The plot of overvoltage against $\log_{10} i$ is a straight line as follows:



The results for a number of cathodes in dil. H_2SO_4 are in harmony with the expectations from the equation. (The value of B in the above eqn. is given as

$$B = \frac{2 \times 2.303 RT}{F}$$

and its value is equal to 0.118 at ordinary temperatures. Its value has been determined experimentally and found to be 0.12)

Importance of overvoltage. (Applications)

① Electroplating of Zn in acidic solutions.

Let us consider the electrolysis of a 0.1M solution of $ZnSO_4$ in contact with 0.1M H_2SO_4 .

The theoretical decomposition potential of H_2 and Zn (from the electrochemical series) are 0.0 volt and -0.76 volt respectively.

Hence H_2 alone should be ~~lib~~ liberated at first at the cathode and only when

all the H^+ ions are reduced as H_2 , zinc should be deposited. But in practice, Zn is also deposited along with the evolution of H_2 at the cathode

The reason is explained as due to overvoltage of H_2 as follows. i.e. The overvoltage of hydrogen (i.e. the actual value - theoretical value) is about +0.57 volt which is very close to the electrode

Potential of Zinc. Hence H_2 is simultaneously evolved along with the deposition of Zinc also. It means that Zn can be electroplated in acidic solutions also.

2) The working of lead accumulators is explained on the basis of overvoltage.

Due to the high overvoltage of Pb metal, during charging Pb is deposited on the cathode instead of H_2 being evolved. This view is confirmed by the fact that if the Pb electrodes are covered with a metal of low overvoltage like platinum etc. H_2 gas is immediately evolved and Pb is not deposited.

3) The principle of overvoltage is used in the industrial production of Cl_2 and NaOH by the electrolysis of a solution of NaCl.

4) By virtue of its position in the electrochemical series, lead should react with dil. HCl. But, this reaction does not occur due to the overvoltage of H_2 . i.e. H_2 is not evolved.

5) The phenomenon of overvoltage is used in several electrolytic processes. If electrodes with high overvoltage are used in electrolytic reductions it amounts to the something as increasing the concentration or activity of hydrogen at the electrodes and thus increasing the reducing power of the electrode.

This fact is ~~made~~ used in electrolytic reductions of organic compounds. Thus electrodes with high overvoltages (e.g. lead electrodes) are used instead of platinum electrodes in some organic electrolytic reductions. i.e. reduction of nitrobenzene.

Nov 2001

What is H_2 overvoltage? mention any two uses (4 marks)

Nov 2002. Define H_2 over potential? mention any two applications of it (4)

POLOROGRAPHY & AMPEROMETRIC TITRATIONS

POLAROGRAPHY

Introduction:-

Polarography is a method of analysis in which the solution to be analysed is electrolysed in such a way that the graph of current against voltage shows, what is the solution and how much present.
(or)

It is instrumental analysis, which consist of measurement of potential difference as current flows in solution and the result obtained can thus be interpreted in terms of nature and conc. of many substances.

Principle:-



* The basic idea is to pass the current between two electrodes.

* One is large in area and other very small.

* Normally both electrodes are of mercury.

* The large electrode is a pool of mercury at the bottom of the cell, at the bottom

* The small electrode is a drop of Hg coming out of very fine capillary tube.

* Thus, if a steady increasing voltage is applied to such a cell, it possible to construct a reproducible current-voltage curve.

* The Current-Voltage Curve is known as polarogram.

* The technique depends upon concentration polarisation produced at the micro electrode.

* It was developed by Czech chemist, Heyrovsky in 1922. He was awarded the chemistry Nobel prize in 1959.

Dropping mercury electrode :- (DME) :-

In this apparatus, a DME which consist of mercury reservoir from which mercury drops down as small drops through a capillary. This acts as cathode and is generally known as indicator or micro electrode.

The anode consist of mercury pool at the bottom of the reservoir, which acts as reference electrode, and its area is larger, so that it is not polarised. Diag

Both cathode and anode are connected across the Battery 'Ba'. 's' is a shunt for adjusting the sensitivity of the galvanometer G. 'p' is the potentiometer.

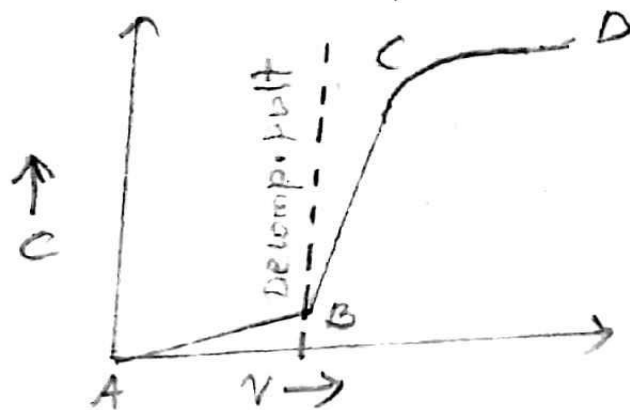
The reservoir is provided for blowing nitrogen gas through the sample, which removes dissolved oxygen from the sample.

Working:-

Consider a polarographic cell, containing a solution of $CdCl_2$, to which an external e.m.f is applied.

The positively charged ions present in the soln will be attracted to the DME.

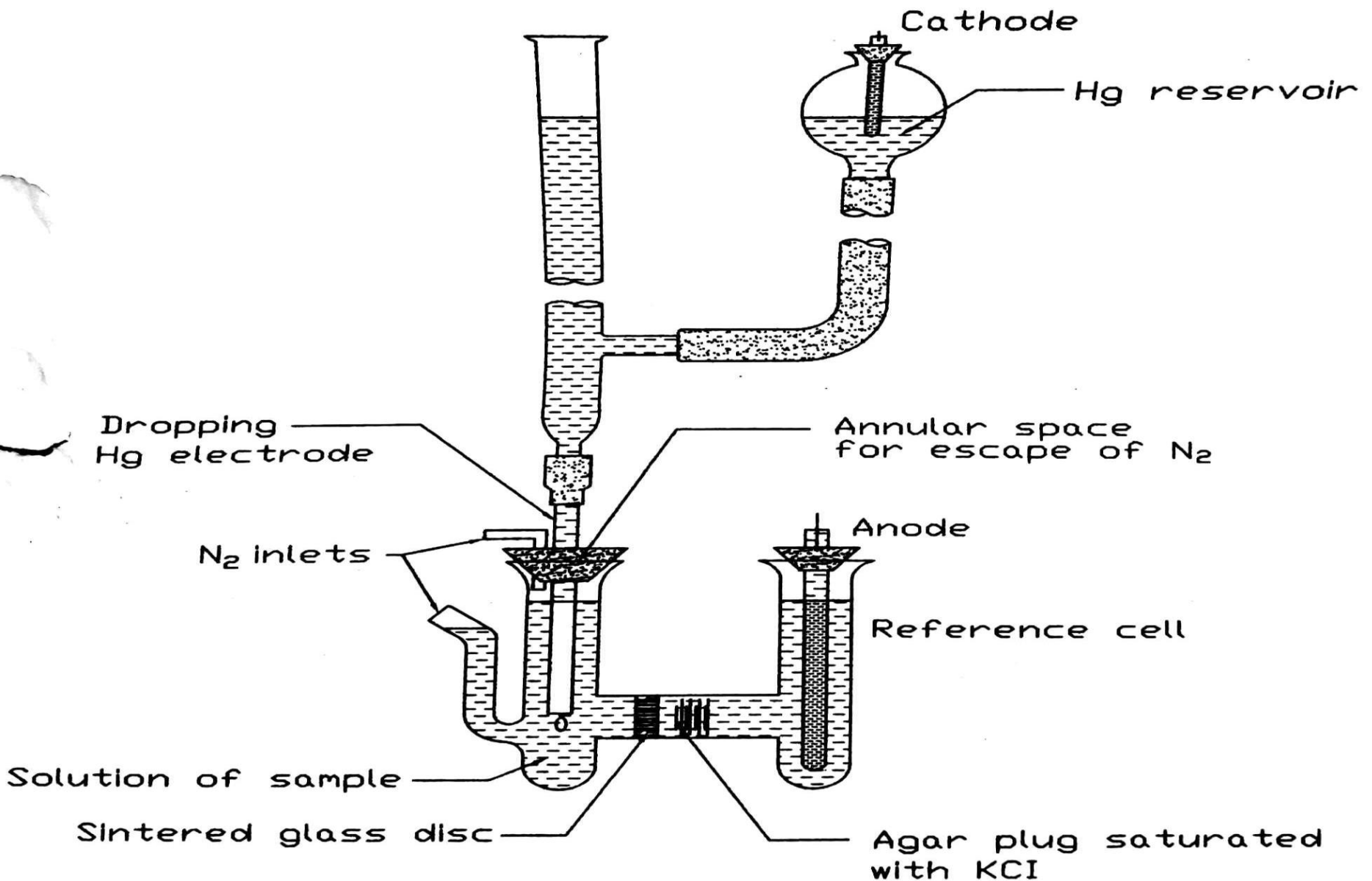
When the applied voltage is increased and the current is recorded, a graph will be obtained



Diffusion current:- From the graph, $A \rightarrow B$, a small current flows, it is residual current.

At 'B', the potential of the electrode becomes equal to the decomposition potential of Cd^{2+} ion.

A dropping mercury electrode and cell



The current then increases along the curve BC. At point C, current no longer increased, but reaches a steady limiting value at point D. Thus current corresponding to the curve 'CD' is known as limiting current.

The difference between the residual current (AB) and the limiting current (CD) is called diffusion current and is generally denoted by i_d

The diffusion current is given by Ilkovic eqn
~~Half wave potential~~:-

$$i_d = 607 n C D^{1/2} m^{2/3} t^{1/6}$$

n = number of electrons transferred per ion.

C = Concentration in millimole/litre

D = Diffusion coefficient in cm^2/sec

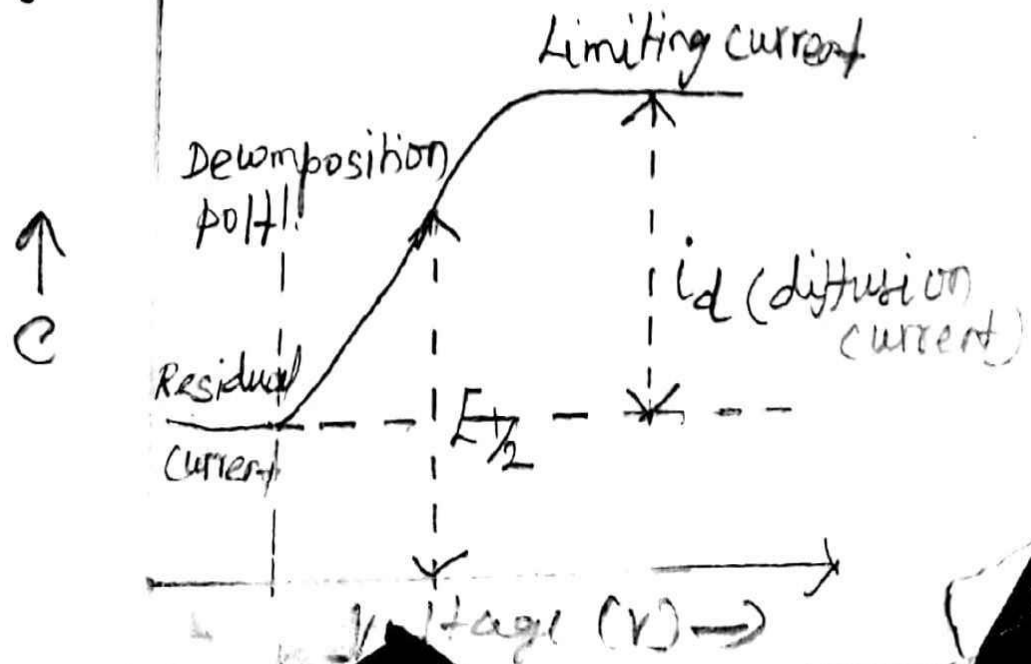
m = mass of Hg flowing per second in mg.

t = drop time in seconds.

from the Ilkovic eqn states that "The observed diffusion current is directly proportional to the concentration of the electroactive material."

Half-wave potential:-

The potential corresponding to the mid point of the diffusion current wave is known as half-wave potential. i.e., half polarographic curve. This is essential for quantitative and qualitative polarographic analysis.



Half-wave potential can understood with the help of reduction rxn occurring in DME.



Nernst eqn for the above eqn (1) is

$$E = E^{\circ} + \frac{0.0591}{n} \log \frac{[\text{Ox}]_0}{[\text{Red}]_0} \quad \text{--- (2)}$$

So the half-wave pot is

$$E_{1/2} = E^{\circ} + \frac{0.0591}{n} \log \left[\frac{D_{\text{red}}}{D_{\text{oxi}}} \right]^{1/2}$$

D_{red} = Diffusion coefficient of the reducible ion

D_{oxi} = Diffusion coefficient of the oxidisable ion

Advantages of D.M.E.

1. Its surface are is reproducible with any given capillary.
2. Electrode surface eliminates poisoning effects.
3. There is a large hydrogen overvoltage on Hg. Therefore reduction of even the alkali metal ions can be observed.
4. Hg forms solid solutions i, amalgams with many metals and hence makes many metal ions more easily reducible.
5. The surface area can be calculated from the weight of the drops.
6. Diffusion current is reproducible
7. A new metal surface is continuously generated, the behaviour of the electrode is thus independent of its past history.

Disadvantages of DME

1. It is easily oxidised. This property severely restricts the use of Hg as an anode.
2. The area of the microelectrode is constant changing the size of the drop of Hg.
3. The capillary may be easily plugged,

Applications of Polarography :- Inorganic :-

1. It is convenient technique for measuring the electrode potentials and for studying electrode rxns.
2. It can be used for the estimations of cations and anions in the presence of interfering ions.

3. Organic applications :-

For the determination of structure, quantitative analysis, identification of compounds is fruitful.

4. For the estimation of sugar
5. Dissolved oxygen in aq. solutions or organic solvent
6. Study of complexes :- composition of complexes.
7. Determination of plant contents :-
8. In medicine :- local anesthetics studied by DME sulphoramide determination.
9. vitamins study.
10. To analyse oxytetracycline and chlorotetracycline.

AMPEROMETRIC TITRATIONS

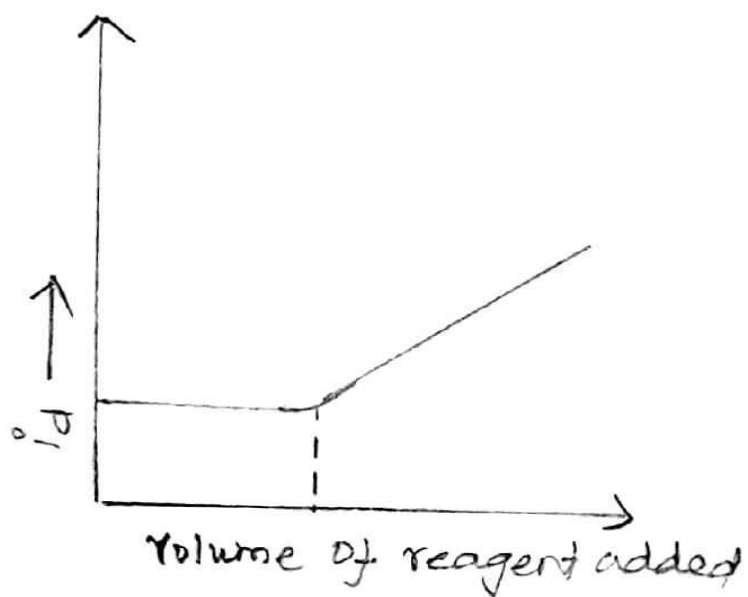
Introduction:-

In amperometric titrations, the voltage applied across the indicator electrode and the reference electrode is kept constant and the diffusion current (which is equal to limiting current - residual current) passing through the cell is measured and plotted against the volume of reagent added.

The amperometric method is considered to be more accurate than polarographic method because it is less dependent upon the characteristic of the capillary and the supporting electrolyte.

Principle:-

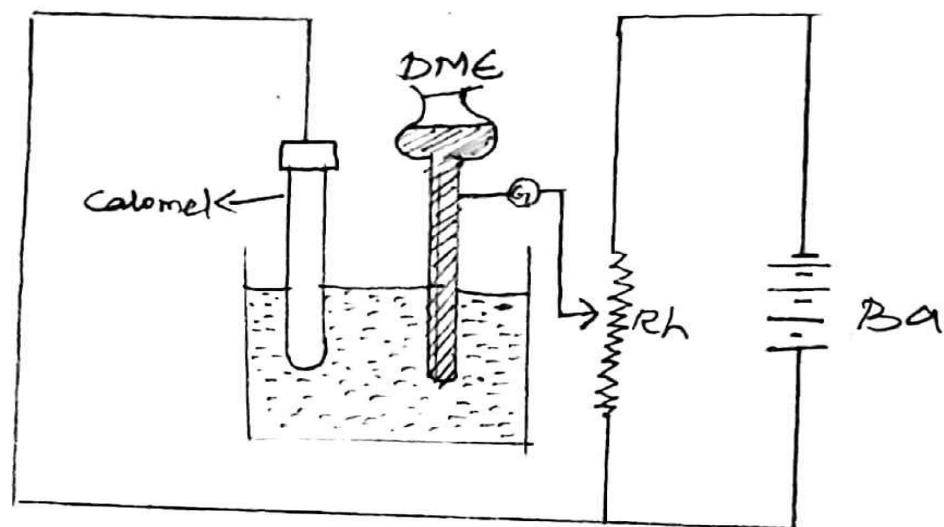
The diffusion current (i_d) will be proportional to the electroactive material in the solution. If the electroactive material is removed, say by interaction with another reagent, the diffusion current is reduced.)



Apparatus:-

The equipment used in amperometric titration is similar to polarography. The titration may be performed either with the DME or rotating platinum electrode.

For DME, the usual polarographic equipment is used with cells modified to allow the addition of titrating reagent.



Calomel electrode is used as a reference electrode. The galvanometer measures the current and the series of rheostat may be used for changing the sensitivity of the galvanometer.

In order to carry out the amperometric titrations, the voltage applied to the indicator electrode and the reference electrode kept constant. The sensitive galvanometer indicates the value of diffusion current after each addition of the titrant.

Advantages of amperometric titrations:-

1. The apparatus used is simple and the characteristics of electrodes are less important.
2. The accuracy is higher than in polarography and the error in this is less.
3. The temperature need not be known provided it remains constant throughout the experiment.
4. The range and the sensitivity of the technique are more than the conductometric titrations.
5. Less conc. range can be measured accurately (0.1 to 0.0001 M - $0.000,001\text{ M}$)
6. The depolarising substances which cannot be determined accurately by polarography, can be successfully determined by this titration.

⑦ Dilute solutions can be titrated with high degree of accuracy.

8. In precipitation titration, it is mostly useful for determine the end point.

Some eg:-

Electrode	Reagent	Substance determined
DME	K_2CrO_4	Pb^{2+} , Ba^{2+}
	$Pb(NO_3)_2$	SO_4^{2-} , F^- , Cl^-
	8-hydroxyquinoline	Mg^{2+} , Zn^{2+} , Cu^{2+}
	DMG	Ni^{2+}
	$K_4Fe(CN)_6$	Zn^{2+}
Rotating platinum	$AgNO_3$	Cl^- , Br^- , I^- , CN^-

Disadvantages:-

1. Inaccurate results are sometimes obtained because of coprecipitation.
2. The foreign substances should not be present in larger concentration than the substance to be titrated. In their presence, the relative current becomes smaller.